

Emily F. Wissel - Curriculum Vitae

EDUCATION

- May 2023 PhD, Emory University, School of Nursing
NSF Graduate Research Fellow (NSF GRFP)
Advisors: Dr. Timothy Read and Dr. Vicki Hertzberg
- 2017 B.S. Psychology, Biology Minor
The University of Texas at Austin

RESEARCH POSITIONS

- 2019 - 2023 PhD Student Researcher
The Microbiome & Maternal Health Outcomes, Emory University
Principal Investigators: Dr. Anne Dunlop and Dr. Tim Read
- 2021 Bioinformatics Intern
Metagenomics & the Canine Microbiome
Hill's Pet Nutrition
- 2018 - 2019 Senior Research Interviewer
Tele-Savvy: A Caregiver Intervention Study, Emory University
Principal Investigator: Dr. Ken Hepburn
- 2018 Research Associate
Population Dynamics of Host-Associated Microbial Communities in *Apis mellifera*, the University of North Carolina in Greensboro
Principal Investigators: Dr. Kasie Raymann & Dr. Louis-Marie Bobay
- 2017 Visiting Research Scientist
Texas Children's Microbiome Center: ASD & the Microbiome, Baylor College of Medicine
Principal Investigator: Dr. Ruth Ann Luna
- 2015 - 2017 Undergraduate Research Assistant, Multiple Labs
The University of Texas at Austin
Principal Investigators: Dr. Adron Harris, Dr. Marci Gleason, Dr. Timothy Loving
- 2015 **Aspiring Scientist Summer Internship Program,**
George Mason University & HHMI Janelia Research Campus
Principal Investigator: Dr. Davi Bock

JOURNAL PUBLICATIONS

Preprints and manuscripts under review are listed above the dashed line

** denotes co-authorship*

Wissel, E. F., Talbot, B. M., Petit, R. A., Hertzberg, V., Dunlop, A., & Read, T. D. (Preprint). hAMRoaster: a tool for comparing performance of AMR gene detection software. *bioRxiv*.

Wissel, E.F. & Risjord, M. (Under Review). Microbiome and the Mind: Is the microbiome part of our cognitive system?

Wissel, E.F., Natarajan Thirulokachander, T., & Kozik, A.J. (Under review, invited submission). Reframing “Healthy” in Vaginal Microbiome Research: a more equitable approach to vaginal microbiome research.

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**Wissel, E.F.**, Leon, L. & Tipton, L. (2022). Opportunities for growth in the growing field of psychobiotics. *Beneficial Microbes*: 0 (0).  
[doi.org/10.3920/BM2022.0051](https://doi.org/10.3920/BM2022.0051)

Ishaq, S. L., **Wissel, E.F.**, Wolf, P., et al. (2022). Designing the Microbes and Social Equity Symposium: A Novel Interdisciplinary Virtual Research Conference Based on Achieving Group-Directed Outputs. *Challenges*, 13(2), 30;  
<https://doi.org/10.3390/challe13020030>

Ishaq, S. L., Parada, F. J., Wolf, P. G., Bonilla, C. Y., Carney, M. A., Benezra, A., **Wissel, E.F.**, ... & Morar, N. (2021). Introducing the microbes and social equity working group: Considering the microbial components of social, environmental, and health justice. *mSystems*, 6(4), e00471-21.

Wright, M. L., Dunlop, A. L., Dunn, A. B., Mitchell, R. M., **Wissel, E. F.**, & Corwin, E. J. (2021). Factors Associated with Vaginal Lactobacillus Predominance Among African American Women Early in Pregnancy. *Journal of Women's Health*.

Rogers AE, Hu Y, Yue Y, **Wissel EF**, Petit III RA, Jarrett S, Christie J, Read TD. (2021). Shiftwork, functional bowel symptoms, and the microbiome. *PeerJ* 9:e11406  
<https://doi.org/10.7717/peerj.11406>

Bobay, L. M., **Wissel, E. F.**, & Raymann, K. (2020). Strain Structure and Dynamics Revealed by Targeted Deep Sequencing of the Honey Bee Gut Microbiome. *mSphere*, 5(4).

Bender, M. E., Read, T. D., Edwards, T. S., Hargita, M., Cutler, A. J., **Wissel, E. F.**, Wise, S. K. (2020). A Comparison of the Bacterial Nasal Microbiome in Allergic Rhinitis Patients and After Immunotherapy. *The Laryngoscope*.

**Wissel, E. F.** & Smith, L. K. (2019). Inter-individual variation shapes the human microbiome. *Brain and Behavioral Sciences*. DOI:  
<https://doi.org/10.1017/S0140525X1800287X>

**Wissel, E. F.\*** & Smith, L. K.\* (2019). Microbes and the mind: How bacteria shape affect, neurological processes, cognition, social relationships, development,

and pathology. Perspectives on Psychological Science. DOI:  
<https://doi.org/10.1177/1745691618809379>

## ORAL PRESENTATIONS

- **Wissel, E. F.** (July 2022). How does the microbiome change during pregnancy? Rapid Fire Talk at SymbNET Summer School.
- Risjord, M. & **Wissel, E. F.** (August 2022). Mind, Body, Spirit...and Poo? Microbiome Research and the Holistic Imperative in Nursing. Panel session “The complexities of 'knowledge' in relation to nursing; philosophical explorations”. IPONS 2022.
- **Wissel, E. F.** (2022 July). Session Chair for the Microbes and Social Equity Virtual Symposium: Blending biological and social (and humanities) writing.
- **Wissel, E. F.**, Holster, S., and Holman, J. on behalf of Ishaq, S. (2022 June). Microbes and Social Equity: what is it and how do we do it? AES Track Hub talk at ASM Microbe 2022.
- **Wissel, E. F.** (2022 April, June). Benchmarking AMR Tools for short-read shotgun metagenome data using simulated metagenomes. Rapid fire oral presentation at ASM MICROBE 2022 and oral presentation at ECCMID 2022.
- **Wissel, E. F.** & Risjord, M. (2022 March). Microbes and the Mind: Is the human microbiome "cognitive"? Presentation to NIH DCEG Microbiome Working Group's journal club.
- **Wissel, E. F.** (2021). Session Chair for Microbes and Social Equity Virtual Symposium: Access to Healthcare.
- **Wissel, E. F.**, Petit, R., Read, T. D., Dunlop, A. L. (2020 March). The Vaginal Virome in Self-Identified African American Women Associated with Bacterial Community during Pregnancy and Preterm Birth. Oral presentation and poster presented at the Keystone Global Virome Conference in Tahoe, CA.
- **Wissel, E.** (2017, August). How to Deal with Missing Data and the Importance of Behavioral Measures. Talk presented to the Baylor College of Medicine Pathology Department.

## AWARDS & GRANTS

- **ECCMID Travel Award.** Declined due to COVID / virtual attendance.
- **NSF Graduate Research Fellowship Program.** Fellowship providing stipend and tuition support for three years. May 2020-2023.
- **RISE Mini-Grant.** The Center for Reproductive Health Research in the Southeast. Emory University. November 2019.
- **Research and Travel Award.** University of North Carolina at Greensboro, Department of Biology. October 2018

- **Award for Excellence in Psychology, Social Sciences, and Human Development:** Poster Presentation, Spring 2017
- **Dean's Honor List**, the University of Texas at Austin: Fall 2014, Fall 2016, Spring 2017, Fall 2017
- **BK Community Scholar Award**, Fall 2014

## POSTERS

- **Wissel, E.** (2022 July). The Emory Pregnancy Project: How the Gut and Vaginal Microbiome Support Healthy Pregnancies. Poster presented at SymbNET Summer School.
- **Wissel, E.**; Raymann, K.; Bobay, L.M. (2018, May). Strain Variance of the Honey Bee Microbiome Across the United States. Poster presented at the NC Biotech Microbiome Symposium.
- **Wissel, E.** (2017, September). Mind-Altering Microbes: How bacteria shape cognition, development, and psychiatric disorders. Talk presented at the Fall Undergraduate Research Fair in Austin, TX.
- **Wissel, E.** & Gleason, M. (2017, April). Sex and the Transition into Parenthood. Poster presented at Spring Undergraduate Research Fair in Austin, TX. (Best Poster Award).
- **Wissel, E.**, Lauritzen, J. S., Spillman, E., Sheridan, A., & Bock, D. (August 2015). The importance of neuronal classification for interpretation of structure-function relationships: multiresolution identification of a Kenyon cell class in the mushroom body of the *Drosophila melanogaster*. Poster presented at ASSIP Conference, Fairfax, VA.

## NON-JOURNAL PUBLICATIONS

- **Wissel, E.F.** (2021) V is for Vagina: Book Chapter in food, feminism, and fermentation 2021 musings. [Available online.](#)
- **Wissel, E.F.** (2021) Addressing Systematic Barriers in Human Microbiome Research. Blog post for the American Society of Microbiology (ASM). ([link](#))
- **Wissel, E.F.** & Guillen, A. (2020, September). The Vaginal Microbiome during Labor and Delivery: A Zine for Healthcare Providers. [Available online.](#)

## WORKSHOPS & TEACHING

- **Foundations for Online Teaching**, Teaching certificate, Summer 2022 at Emory University.
- **Research Ethics, Literature Searches, and Synthesizing Theory.** Guest Lectures for The Laney Summer Opportunity for Advanced Research intensive course, June 2022.
- **The Human Vaginal Microbiome**, Guest Lecture for Intro to Microbiology course at New College of Florida, April 2022

- **Foundations of Microbiology for Nurses** - Teaching Associate, Spring 2021 at Emory University
- **Palliative Care** - Teaching Assistant, Fall 2020 at Emory University
- **Introduction to R** - Workshop host, Summer 2018 at University of North Carolina in Greensboro
- **Creating Publication-Ready Figures in R** - Workshop host, Summer 2018 at University of North Carolina in Greensboro
- **Introductory Biology Lab** - Teaching Assistant, Fall 2018 at the University of North Carolina in Greensboro

## WORKSHOPS ATTENDED

- **SymbNET Summer School 2022:** This Summer School is designed for PhD students interested in host-microbe symbioses to acquire an in-depth understanding of the field from diverse perspectives through the interaction with experts in this research area.
- **Open Science Grid 2018:** training for using high-throughput computing (HTC) systems, including the HTCondor cluster.

## SERVICE & MENTORSHIP

- **Microbes and Social Equity Working Group:** I have been on the planning committee for three virtual conferences and two virtual speaker series with MSE. I am currently the Director of Resource Dissemination for MSE and help manage their twitter account.
- **NSURP (2020, 2022):** National Summer Undergraduate Research Program. Virtually mentored two students on a coding project and two on manuscript development.
- **Pipeline Emory (2020):** I serve as a mentor to five undergraduate and five high school students, facilitate weekly discussions, and run interactive educational activities focused on public health and academic readiness.
- **Data Science Mentor (2018):** I spent several hours each week meeting with first generation college students in their final year of college. I helped them navigate their career goals and options. We reviewed principles of data visualization in R and data science in Python

## PRESS

- **Brains and Bacteria.** Cover Story in The Observer covering publication in Perspectives in Psychological Science. ([link](#)).
- **I Study Poop: Adventures from Scientists on the Front Line.** Online interview for Dr. Callaghan's microbiome blog ([link](#)).
- **Science Sucks Podcast.** Earth Day Special Extravaganza ([link](#)).